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WHITING SCHOOL
of ENGINEERING

FPGA Design Using VHDL

Module 1D: Example of Entity/Architecture Pair

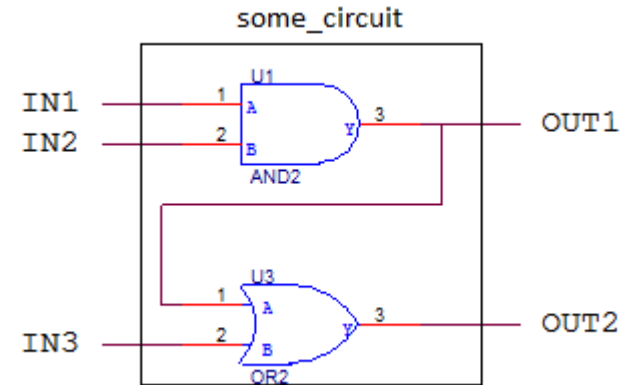
Ex 1: Entity/Architecture Pair

```
-- VHDL file saved as some_circuit.vhd
library IEEE;
use IEEE.std_logic_1164.all;
use work.all;
```

```
entity some_circuit is port (
    in1, in2, in3 : in std_logic;
    out1, out2 : out std_logic);
end some_circuit;
```

```
architecture arch of some_circuit is
begin
    out1 <= in1 and in2;
    out2 <= out1 or in3;
end arch;
```

Note: <= is a “signal assignment”



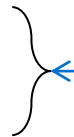
- Quiz Style Question
 - How would you implement the above circuit using correct VHDL syntax to avoid this error?

Ex2: Entity/Architecture Pair

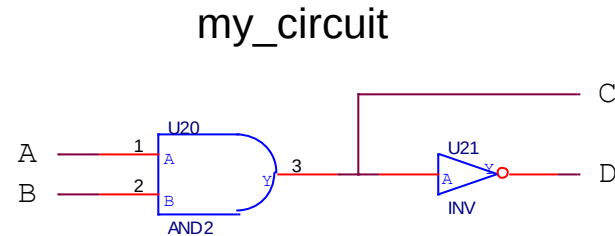
```
library IEEE;
use IEEE.std_logic_1164.all;
use work.all;
```

```
entity my_circuit is port (
    a, b: in std_logic;
    c,d : out std_logic);
end my_circuit;
```

```
architecture behavioral of my_circuit is
    signal andab : std_logic;
begin
    c <= andab;
    d <= not andab;
    andab <= a and b;
    -- These are concurrent assignments...
    -- all happen at the same time
end behavioral;
```



This is known as concurrent
signal assignment (CSA)
More on that later



- Note that this is the preferred solution to problem mentioned previous slide